

Jesús García Sánchez

PhD Candidate. Physicist. Biomathematician



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Spanish (Native)

English (Bilingual)

French (Working Proficiency)

Professional Experience

Mar. 2025 - Now	Predoctoral Research Fellow at University of Granada.
Nov. 2023 - Mar. 2025	Project Research Associate at University of Granada.
Jan. 2023 - Nov. 2023	Data Scientist at Eurofins Cavendish.
Sep. 2022 - Jan. 2023	Junior Programmer at T-Systems Iberia.
Jan. 2020 - Dec. 2020	Educational Activities Technician Internship at Parque de las Ciencias Granada.

Education

2024	MSc in Physics and Mathematics. Specialization in Biomathematics. University of Granada
2022	BSc in Physics. University of Granada

Research Articles

2025	Fernandez-Carabantes, J., Titos, M., D'Auria, L., Garcia, J., Garcia, L., Benitez, C. RNN-DAS: A New Deep Learning Approach for Detection and Real-Time Monitoring of Volcano-Tectonic Events Using Distributed Acoustic Sensing. Journal of Geophysical Research: Solid Earth, 130, e2025JB031756, 2025. (ISI'23): 4.24 (Q1 – Geophysics). Link
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Research Projects

2025 - Now	Multi-scale Spatio-Temporal Analysis of Research Data (MuSTARD). Reference: PID2023-152301OB-I00. Link
2023 - 2025	DigiVolCan - una infraestructura digital para la previsión de erupciones volcánicas en Canarias. Reference: PLEC2022-009271. Link

2023 Investigación y desarrollo de elementos específicos para la mitigación de ruido submarino generado por obras portuarias y dragados. (CDTI project).

International Workshops and Conferences

- 2026 García Sánchez, J., Benítez, C., D'Auria, L., García, L., Camacho, J. **Multivariate Statistical analysis of DAS seismic data.** Galileo conference: Fibre Optic Sensing in Geosciences. [Link](#)
- 2026 Fernández Carrascosa, L., García Sánchez, J., Fakhruz I., Camacho J., García L.. **Monitoring urban traffic with Distributed Acoustic Sensing and ANOVA Simultaneous Component Analysis.** Galileo conference: Fibre Optic Sensing in Geosciences . [Link](#)
- 2026 García Sánchez J., , SoroChan Armstrong M. , García L., Camacho J. **Analysis Optimization of Chemical Separations Data with Time-Frequency Representations.** XX Chemometrics in Analytical Chemistry International Conference.
- 2025 **Three Minute Thesis (3MT) Competition Winner** (1st place out of 101 participants) at IEEE International Geoscience and Remote Sensing Symposium 2025, Brisbane (Australia). [Video](#), [Link](#)
- 2025 García Sánchez, J., García, L., D'Auria, L., Fernández-Carabantes, J., Benítez Ortúzar, C., Camacho, J. **Volcanic eruption forecast using PCA.** IEEE International Geoscience and Remote Sensing Symposium 2025, Brisbane (Australia). Poster. [Link](#)
- 2025 Fernández-Carabantes, J. and Titos, M. and D'Auria, L. and García, J. and García, L. and Benítez, C. **RNN-DAS: A New Deep Learning Approach for Detection and Real-Time Monitoring of Volcano-Tectonic Events Using Distributed Acoustic Sensing.** IEEE International Geoscience and Remote Sensing Symposium 2025, Brisbane (Australia). Oral presentation.
- 2025 **Chair** at IEEE International Geoscience and Remote Sensing Symposium 2025, Brisbane (Australia).
- 2025 García Sánchez, J., Camacho, J., García, L., SoroChan Armstrong, M. **Multivariate exploratory data analysis of spatio-temporal data using data fusion, FFT and component models.** Spatial Statistics 2025: At the Dawn of AI, Noordwijk (The Netherlands). Poster. [Link](#)
- 2024 García, J., Camacho, J., Dey, P.S., Fakhruzi, I., Benítez, C., Barrancos, J., D'Auria, L., García, L., **Exploratory Data Analysis of Seismic Data using Distributed Acoustic Sensing and Principal Component Analysis.** 9th International Conference in Spectral Imaging-IASIM, Bilbao (Spain). Poster.

National Workshops and Conferences

2025 García-Sánchez, J., Camacho, J., García L., Benítez-Ortúzar, C., D'Auria, L. **Forecasting volcanic eruptions using interpretable multivariate analysis**. CITIC-Coffee 2025, Granada. Poster. [Link](#)

Awards

2025 **Excellence in Technical Communication** Student Prize Award (3MT First Place out of 101 participants), IEEE Geoscience and Remote Sensing Society. [Link](#)

Teaching and Supervisions

2025 Teaching: Network Fundamentals. Bachelor in Computer Engineering. University of Granada.

2025 Supervision: Bachelor in Computer Engineering and Mathematics' Thesis: "Análisis Multivariante de Datos Sísmicos" by Santiago Hernández Hernández. University of Granada.

2023 Online course: Introducción a la programación como herramienta. Recursos para el uso de Python. #YoSigoPublicando. Universidad de Granada [Link](#)

Software

2025 MSPC-Seismic: DigiVolCan's real time monitoring and diagnosis system of seismic signals. [Link](#)

Fellowships

Mar. 2025 - Now **Predoctoral Research Fellow**. Project (FPI-funded): Multi-scale Spatio-Temporal Analysis of Research Data (MuSTARD). Reference: PID2023-152301OB-I00. University of Granada.

Science Outreach and Public Engagement

Sep. 2025 **Workshop: "Decisiones espacio-temporales"** at European Researchers' Night, Granada. [Link](#)

Sep. 2025 **Talk: "¿Cuándo falla la ciencia?"** at European Researchers' Night, Granada. [Link](#)

Apr. 2025 **Workshop: "Decisiones espacio-temporales"** at Feria de las Ingenierías ETSIT, Universidad de Granada. [Link](#)

